



**Caringbah High School**  
 Year 12 2025  
**Mathematics Extension 2**  
 HSC Course  
 Assessment Task 4 – Trial HSC Examination

**General Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- NESA-approved calculators may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

**Total marks – 100**

**Section I**      **10 marks**

Attempt Questions 1-10  
 Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

**Section II**      **90 marks**

Attempt Questions 11-16  
 Write your answers in the answer booklets provided. Ensure your name or student number is clearly visible.

**Name:** \_\_\_\_\_

**Class:** \_\_\_\_\_

| Marker's Use Only |            |     |     |     |     |     |       |
|-------------------|------------|-----|-----|-----|-----|-----|-------|
| Section I         | Section II |     |     |     |     |     | Total |
| Q 1-10            | Q11        | Q12 | Q13 | Q14 | Q15 | Q16 |       |
| /10               | /15        | /15 | /15 | /15 | /15 | /15 | /100  |

## Section I

**10 marks**

**Attempt Questions 1-10**

**Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10

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1. If  $\frac{1}{x^2 + x} = \frac{A}{x} + \frac{B}{x + 1}$ , what are the values of  $A$  and  $B$ ?

(A)  $A = -1, B = -1$

(B)  $A = 1, B = -1$

(C)  $A = -1, B = 1$

(D)  $A = 1, B = 1$

2. If  $z = \frac{3 + 4i}{1 + 2i}$ , the imaginary component of  $z$  is:

(A)  $-2$

(B)  $\frac{2}{5}$

(C)  $-\frac{2}{5}$

(D)  $2$

3. Given the complex number  $z = a + ib$ , where  $a, b \in \mathbb{R}$ ,  $a \neq 0$ ,

$\frac{4z\bar{z}}{(z+\bar{z})^2}$  is equivalent to:

(A)  $4[Re(z) \times Im(z)]$

(B)  $1 + \left(\frac{Im(z)}{Re(z)}\right)^2$

(C)  $4([Re(z)]^2 + [Im(z)]^2)$

(D)  $\frac{2 \times Im(z)}{[Re(z)]^2}$

4. Consider the following statement.

“If I live in Berlin then I live in Germany.”

What is the negation of this statement?

(A) If I don't live in Berlin then I don't live in Germany.

(B) If I don't live in Germany then I don't live in Berlin.

(C) If I live in Germany then I live in Berlin.

(D) I live in Berlin and I don't live in Germany.

5. The value of  $i + i^2 + i^3 + \dots + i^{2025}$  is:

(A)  $i$

(B)  $-1$

(C)  $1$

(D)  $-i$

6. Which of the following is the vector equation of a line that passes through the points  $(8, -2, 8)$  and  $(0, 7, 0)$ ?

(A) 
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 5 \\ 8 \end{pmatrix}$$

(B) 
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 5 \\ 8 \end{pmatrix}$$

(C) 
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ -9 \\ 8 \end{pmatrix}$$

(D) 
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 8 \\ -2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -8 \\ -9 \\ -8 \end{pmatrix}$$

7. A variable force acts on a particle, causing it to move in a straight line. At time  $t$  seconds, where  $t \geq 0$ , its velocity  $v$  metres per second and position  $x$  metres from the origin are such that  $v = e^x \sin x$ .

The acceleration of the particle, in  $\text{ms}^{-2}$ , can be expressed as:

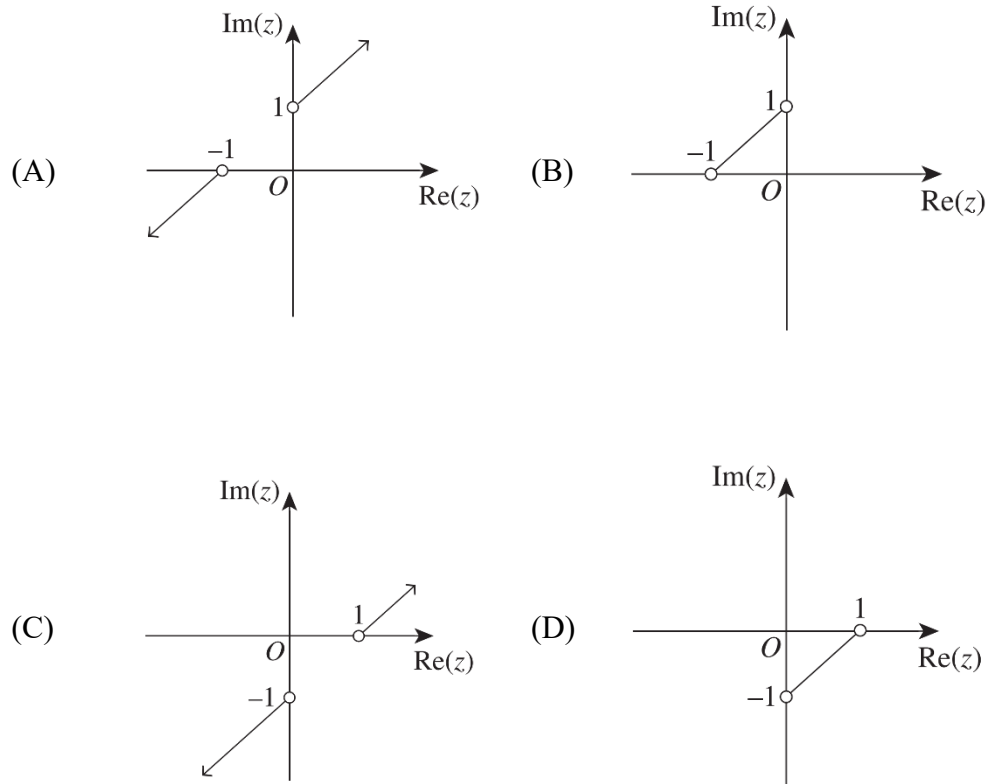
(A)  $e^x(\sin x + \cos x)$

(B)  $e^x \sin x (\sin x + \cos x)$

(C)  $\frac{1}{2} e^{2x} \sin^2 x$

(D)  $e^{2x} \left( \sin^2 x + \frac{1}{2} \sin 2x \right)$

8. Which of the following Argand diagrams represents the solutions to the equation  $\text{Arg}(z - i) = \text{Arg}(z + 1)$ ?



9. Let  $\underline{a} = \underline{i} + \underline{j}$ ,  $\underline{b} = \underline{i} - \underline{j}$  and  $\underline{c} = \underline{i} + 2\underline{j} + 3\underline{k}$ .

If  $\underline{n}$  is a unit vector such that  $\underline{a} \cdot \underline{n} = 0$  and  $\underline{b} \cdot \underline{n} = 0$ , then  $|\underline{c} \cdot \underline{n}|$  is equal to:

- (A) 2  
(B) 3  
(C) 4  
(D) 5

10. Which expression is equal to

$$\int_0^a [f(a-x) + f(a+x)] dx \quad ?$$

(A)  $\int_0^a f(x) dx$

(B)  $\int_0^{2a} f(x) dx$

(C)  $2 \int_0^a f(x) dx$

(D)  $\int_{-a}^a f(x) dx$

**End of Section I**

## Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

(a) If  $z = \cos\theta + i\sin\theta$ , simplify  $z + \frac{1}{z}$ . 1

(b) Given that  $\underline{a} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$  and  $\underline{b} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ , find the angle between  $\underline{a}$  and  $\underline{b}$  2  
to the nearest degree.

(c) Sketch the region defined by  $|z + 1| \leq 2$  and  $\text{Im}(z) \geq 0$ . 2

(d) Find  $\int (x^2 + 6x + 13)^{-1} dx$  2

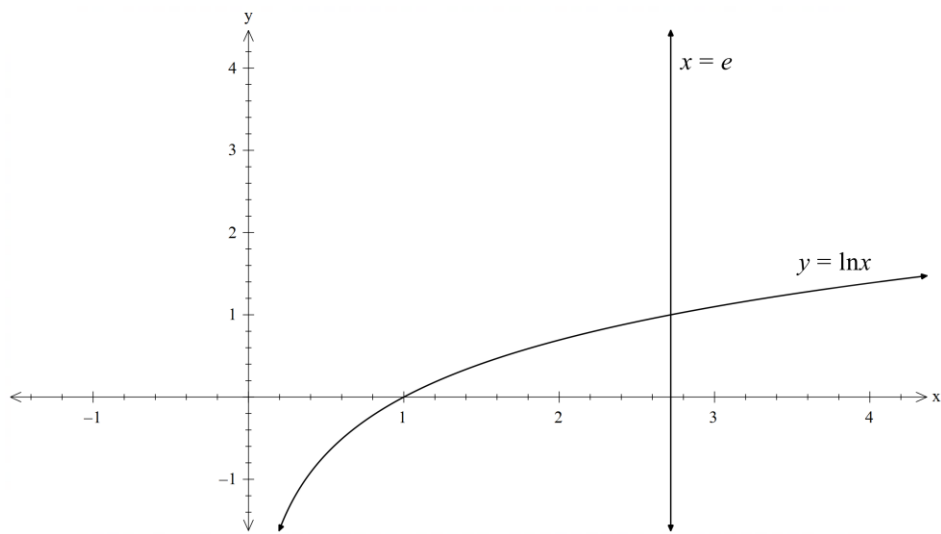
(e) (i) Find the square roots of  $3 + 4i$ . 2

(ii) Hence, or otherwise, solve the equation 2

$$z^2 + iz - 1 - i = 0$$

Question 11 continues on page 9

- (f) The region  $R$  is bounded by the graph of  $y = \ln x$ , the line  $x = e$  and the  $x$ -axis. 4



Find the **volume** of the solid of revolution formed when the region  $R$  is rotated about the  $x$ -axis.

**End of Question 11**

**Question 12** (15 marks) Use a SEPARATE writing booklet.

- (a) By using a suitable substitution, or otherwise, find 2

$$\int \frac{\sin^3 x}{\cos^2 x} dx$$

- (b) A body moves in a straight line so that when its displacement from a fixed origin  $O$  is  $x$  metres, its acceleration,  $a$ , is  $-4x \text{ ms}^{-2}$ . 3

The body accelerates from rest and its velocity,  $v$ , is equal to  $-2 \text{ ms}^{-1}$  as it passes through the origin. The body then comes to rest again.

Find  $v$  in terms of  $x$  for this interval.

- (c) Prove by contradiction that  $\sqrt{11} - \sqrt{5} < \sqrt{2}$ . 2

- (d) Evaluate  $\int_0^6 \sqrt{36 - x^2} dx$  2

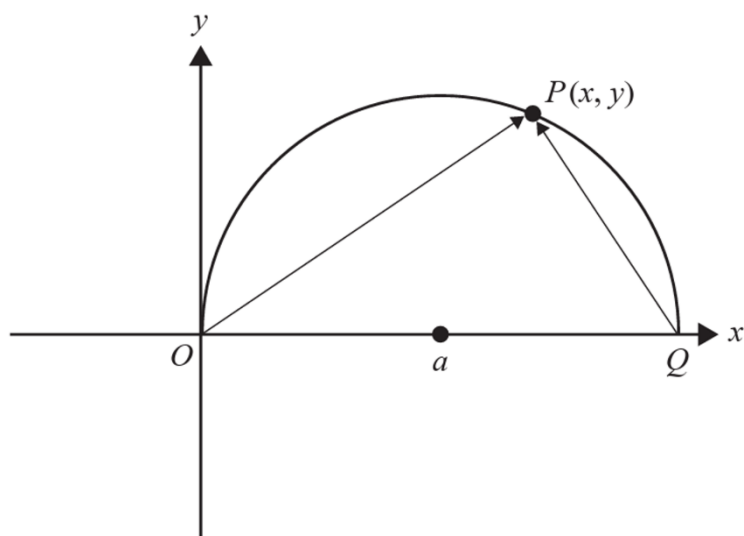
- (e) The point  $P$  represents the complex number  $z$  in the complex plane. 3

If  $z = x + iy$ , prove that  $\operatorname{Re}\left(\frac{z - ib}{z - a}\right) = 0$  is the equation of a circle,

where  $a$  and  $b$  are real, and state its centre and radius in terms of  $a$  and  $b$ .

**Question 12 continues on page 11**

- (f)  $OPQ$  is a semi-circle of radius  $a$  with equation  $y = \sqrt{a^2 - (x - a)^2}$ .  
 $P(x, y)$  is a point on the semi-circle  $OPQ$ , as shown below.



- (i) Express  $\overrightarrow{OP}$  and  $\overrightarrow{QP}$  in terms of  $a, x, y, \underline{i}$  and  $\underline{j}$ . 1
- (ii) Hence determine whether  $\overrightarrow{OP}$  is perpendicular to  $\overrightarrow{QP}$ . 2

**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet.

- (a) A particle is moving in a straight line and its position at time  $t$  is given by

$$x = 2 + \sin 2t + \sqrt{3} \cos 2t$$

- (i) Prove that the particle is undergoing simple harmonic motion about  $x = 2$ . 2
- (ii) Find the amplitude of motion. 1
- (iii) When does the particle first reach its maximum speed after time  $t = 0$ ? 2
- (b) Show that  $\frac{a^4 + a^2 + 1}{a^2} \geq 3$  for all  $a$ . 1
- (c) (i) Show that  $\left(z - \frac{1}{z}\right)^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$  2
- (ii) By letting  $z = \cos\theta + i\sin\theta$ , or otherwise, show that 2
- $$32\sin^5\theta = 2\sin 5\theta - 10\sin 3\theta + 20\sin\theta$$
- (iii) Hence, find the exact value of  $\int_0^{\frac{\pi}{2}} \sin^5\theta \, d\theta$  2
- (d) Prove that the product of any three consecutive numbers is even. 3

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

(a) The line  $l$  has equation  $\underline{r} = \begin{pmatrix} 0 \\ 3 \\ -5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ b \\ 2 \end{pmatrix},$

where  $\mu$  is a parameter and  $b$  is a constant.

The points  $P$  and  $Q$  have coordinates  $(-1, 2, 3)$  and  $(-2, 4, 2)$  respectively.

(i) Find a vector equation of the line  $PQ$ . 1

Leave your answer in the form of  $\underline{r} = \underline{a} + \lambda \underline{b}$ .

(ii) Find the value of  $b$  for which the acute angle between the line  $l$  and 2  
the line  $PQ$  is  $\cos^{-1}\left(\frac{1}{6}\right)$ .

(b) Consider the following proposition:

Suppose  $x, y \in \mathbb{R}$ . “If  $y^3 + yx^2 \leq x^3 + xy^2$ , then  $y \leq x$ .”

(i) State the contrapositive of this proposition. 1

(ii) Hence, prove the above proposition is true. 2

(c) Two lines  $\underline{r}_1$  and  $\underline{r}_2$  have equations

$$\underline{r}_1 = \begin{pmatrix} 0 \\ 5 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} \text{ and } \underline{r}_2 = \begin{pmatrix} -2 \\ 4 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} \text{ where } \lambda, \mu \in \mathbb{R}$$

The point  $A$  lies on  $\underline{r}_1$  with parameter  $\lambda = p$ , and the point  $B$  lies on  $\underline{r}_2$  with parameter  $\mu = q$ .

(i) Show that  $\overrightarrow{AB}$  can be expressed as the column vector 1

$$\begin{pmatrix} -2 - q + p \\ -1 + 2q - 4p \\ -3 + 2q - 3p \end{pmatrix}$$

(ii) Find the value of  $|\overrightarrow{AB}|$  when  $\overrightarrow{AB}$  is perpendicular to both  $\underline{r}_1$  and  $\underline{r}_2$ . 3

**Question 14 continues on page 14**

- (d) Consider the spheres  $S_1$  and  $S_2$ . 2

$$S_1 : x^2 - 2x + y^2 + 4y + z^2 - 4z = 0$$

$$S_2 : x^2 + y^2 + z^2 + 10x + 2z + 10 = 0$$

Show that  $S_1$  and  $S_2$  intersect.

- (e) Find 3

$$\int \frac{e^x}{e^{2x} + 3e^x + 2} dx$$

**End of Question 14**

**Question 15** (15 marks) Use a SEPARATE writing booklet.

(a) Find

2

$$\int (1 + 2x^2)e^{x^2} dx$$

(b) (i) Show that  $|\sin(\alpha + \beta)| \leq |\sin\alpha| + |\sin\beta|$

1

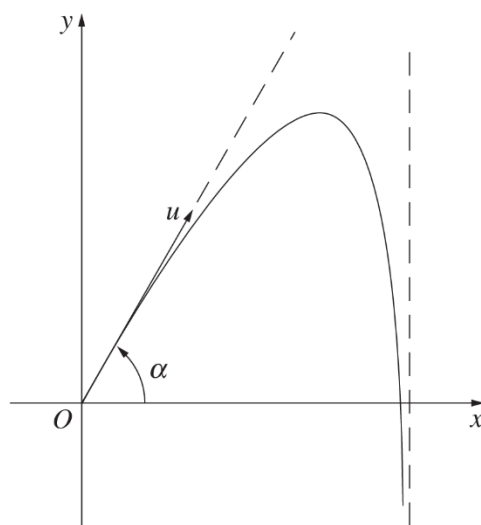
(ii) Under what conditions will equality hold in part (i)?

1

(c) A particle of mass  $m$  is thrown from the top,  $O$ , of a very tall building with an initial velocity  $u$  at an angle  $\alpha$  to the horizontal. The particle experiences the effect of gravity, and a resistance proportional to its velocity in both the horizontal and vertical directions. The equations of motion in the horizontal and vertical directions are given respectively by

$$\ddot{x} = -k\dot{x} \quad \text{and} \quad \ddot{y} = -k\dot{y} - g,$$

where  $k$  is a constant and the acceleration due to gravity is  $g$ .  
(You are not required to show these.)



(i) Derive the result  $\dot{x} = ue^{-kt} \cos\alpha$  from the relevant equation of motion.

2

(ii) Verify that  $\dot{y} = \frac{1}{k} ((k u \sin\alpha + g)e^{-kt} - g)$  satisfies the appropriate equation of motion.

2

(iii) Find the value of  $t$  when the particle reaches its maximum height.

1

(iv) What is the limiting value of the horizontal displacement of the particle?

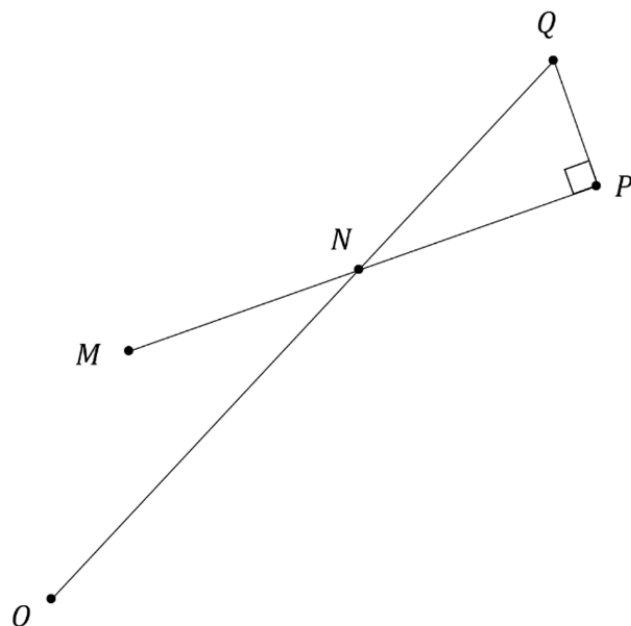
2

**Question 15 continues on page 16**

- (d)  $M$  and  $P$  are two points with position vectors  $\overrightarrow{OM} = \underline{m}$  and  $\overrightarrow{OP} = \underline{p}$  4

respectively.  $N$  is the midpoint of  $MP$  and has position vector  $\overrightarrow{ON} = \underline{n}$ .

$PQ$  is perpendicular to  $MN$  and  $\overrightarrow{NQ} = \lambda \overrightarrow{ON}$ .



NOT TO  
SCALE

Show that

$$\lambda = \frac{|\underline{m}|^2 + |\underline{n}|^2 - 2 \underline{m} \cdot \underline{n}}{|\underline{n}|^2 - \underline{m} \cdot \underline{n}}$$

**End of Question 15**

**Question 16** (15 marks) Use a SEPARATE writing booklet.

- (a) A particle of unit mass is allowed to fall under gravity from rest in a medium which exerts a resistance proportional to the speed,  $v$ , of the particle.

- (i) Show that the particle reaches a terminal velocity  $T$ , 1

given by  $T = \frac{g}{k}$  where  $k$  is a positive constant.

- (ii) Show that the distance fallen to reach half of its terminal velocity,  $\frac{T}{2}$  4

is given by

$$x = \frac{T^2}{g} \ln(2) - \frac{T^2}{2g}$$

- (b) Let  $P(x)$  be a monic polynomial of degree  $n$  with rational coefficients and 2  
zeroes  $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ .

Show that

$$P'(x) = P(x) \left( \frac{1}{x - \alpha_1} + \frac{1}{x - \alpha_2} + \frac{1}{x - \alpha_3} + \dots + \frac{1}{x - \alpha_n} \right)$$

**Question 16 continues on page 18**

- (c) (i) Given that for  $k > 0$ ,  $2k + 3 > 2\sqrt{(k + 1)(k + 2)}$ , 4

use mathematical induction to show that

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$$

for all positive integers  $n$ .

- (ii) Use the graph of  $y = \frac{1}{\sqrt{x}}$  to show that 2

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

- (iii) Hence show that 2

$$198 < \sum_{r=1}^{10\,000} \frac{1}{\sqrt{r}} < 199$$

**End of Examination**

# 2025 Extension 2 Trial Solutions

## Section 1

$$Q1) \frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$

$$1 = A(x+1) + Bx$$

$$\text{Let } x=0, \quad x=-1, \\ A=1 \quad B=-1$$

B.

- Q1) B  
Q2) C  
Q3) B  
Q4) D  
Q5) A  
Q6) C  
Q7) D  
Q8) A  
Q9) B  
Q10) B

$$Q2) z = \frac{3+4i}{1+2i} \times \frac{1-2i}{1-2i}$$

$$= \frac{3-6i+4i+8}{1+4}$$

$$= \frac{11-2i}{5}$$

$$\therefore \operatorname{Im}(z) = -\frac{2}{5}$$

C.

$$Q3) \frac{4z\bar{z}}{(z+\bar{z})^2} = \frac{4(a^2+b^2)}{4a^2}$$

$$= \frac{a^2+b^2}{a^2}$$

$$= 1 + \left(\frac{b}{a}\right)^2$$

$$1 + \left(\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)}\right)^2 = 1 + \left(\frac{b}{a}\right)^2$$

B.

Q4) To negate a statement of the form "If P then Q," the negation is "P and not Q."

- A. Common error
- B. Contrapositive
- C. Converse
- D. Negation

D.

$$Q5) \quad i + i^2 + i^3 + \dots + i^{2024} = 0$$

$$i + i^2 + i^3 + \dots + i^{2024} + i^{2025}$$

$$= 0 + i^{2025}$$

$$= 0 + i$$

$$= i$$

A.

$$Q6) \quad \vec{r} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} + \lambda \vec{BA}$$

$$\vec{r} = \begin{pmatrix} 0 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ -9 \\ 8 \end{pmatrix}$$

C.

$$Q7) \quad v = e^x \sin x$$

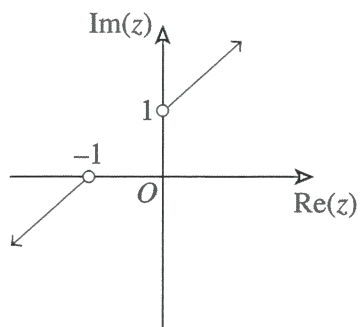
$$\frac{dv}{dx} = e^x \sin x + e^x \cos x$$

$$v \frac{dv}{dx} = e^x \sin x \times e^x (\sin x + \cos x)$$

$$a = e^{2x} \left( \sin^2 x + \frac{1}{2} \sin 2x \right)$$

D.

Q8)



A.

Q9) let  $\underline{a} = x\underline{i} + y\underline{j} + z\underline{k}$

$$|\underline{a}| = x^2 + y^2 + z^2 = 1$$

$$\underline{a} \cdot \underline{a} = 0$$

$$x + y = 0$$

$$\underline{b} \cdot \underline{a} = 0$$

$$x - y = 0$$

$$x = y = 0$$

$$\therefore \underline{a} = z\underline{k}$$

$$\underline{c} \cdot \underline{a} = 3z$$

$$|\underline{c} \cdot \underline{a}| = 3$$

B.

Q10)  $\int_0^a f(a-x) + f(a+x) dx$

$$= \int_0^a f(a-x) dx + \int_0^a f(a+x) dx$$

$$= \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

$$= \int_0^{2a} f(x) dx$$

B.

## Section 2

Q11)a)

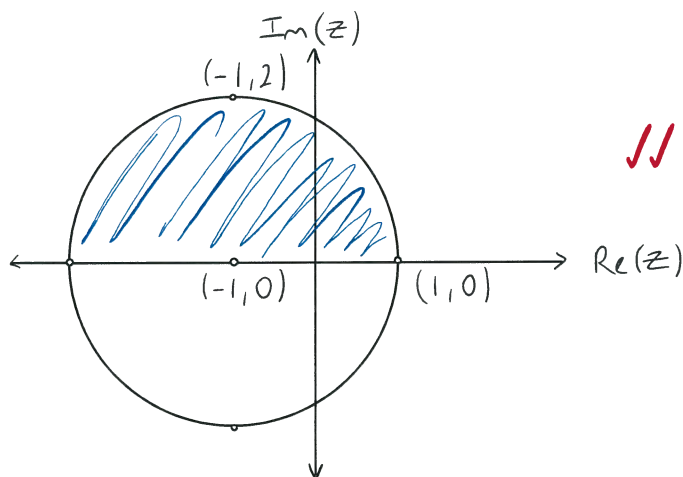
$$\begin{aligned} z + \frac{1}{z} &= \cos\theta + i\sin\theta + (\cos\theta + i\sin\theta)^{-1} \\ &= \cos\theta + i\sin\theta + \cos(-\theta) + i\sin(-\theta) \quad (\text{De Moivre}) \\ &= \cos\theta + i\sin\theta + \cos\theta - i\sin\theta \\ &= 2\cos\theta \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{b) } \cos\theta &= \frac{a \cdot b}{|a||b|} \\ &= \frac{2+2-2}{\sqrt{6} \times \sqrt{9}} \\ &= \frac{2}{\sqrt{54}} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \theta &= \cos^{-1}\left(\frac{2}{\sqrt{54}}\right) \\ &= 74^\circ \quad \checkmark \end{aligned}$$

$$\text{c) } |z - (-1)| \leq 2$$

circle with centre  $(-1, 0)$  and radius 2.



$$\begin{aligned}
 d) \quad & \int \frac{1}{x^2 + 6x + 13} dx \\
 &= \int \frac{1}{x^2 + 6x + 9 + 13 - 9} dx \\
 &= \int \frac{1}{(x+3)^2 + 4} dx \quad \checkmark \\
 &= \frac{1}{2} \tan^{-1} \left( \frac{x+3}{2} \right) + c \quad \checkmark
 \end{aligned}$$

$$e) i) (x+iy)^2 = 3+4i$$

$$x^2 - y^2 = 3 \quad (1)$$

$$x^2 + y^2 = 5 \quad (2)$$

$$(1) + (2),$$

$$(1) - (2),$$

$$\begin{aligned}
 2x^2 &= 8 \\
 x &= \pm 2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 2y^2 &= 2 \\
 y &= \pm 1
 \end{aligned}$$

Since  $\text{Im}(3+4i) > 0$

$2+i$  and  $-2-i$  are the square roots.  $\checkmark$

$$ii) z^2 + iz - (1+i) = 0$$

$$z = \frac{-i \pm \sqrt{i^2 - 4 \times 1 \times -(1+i)}}{2}$$

$$z = \frac{-i \pm \sqrt{3+4i}}{2} \quad \checkmark$$

$$z = \frac{-i}{2} \pm \frac{2+i}{2} \quad \text{or} \quad \frac{-i}{2} \pm \frac{(-2-i)}{2}$$

$$= \frac{-i}{2} + 1 + \frac{i}{2}, \quad \frac{-i}{2} - 1 - \frac{i}{2}$$

$$= 1, -1-i \quad \checkmark$$

duplicate solutions

$$f) \quad V = \pi \int_1^e (\ln x)^2 dx$$

$$u = (\ln x)^2$$

$$u' = 2 \ln x \times \frac{1}{x}$$

$$v' = 1$$

$$v = x$$

$$V = \pi \left[ x (\ln x)^2 \right]_1^e - 2\pi \int_1^e \ln x dx \quad \checkmark$$

$$u = \ln x$$

$$u' = \frac{1}{x}$$

$$v' = 1$$

$$v = x$$

$$V = \pi(e - 0) - 2\pi \left[ x \ln x \right]_1^e - \int_1^e 1 dx \quad \checkmark$$

$$\begin{aligned} V &= \pi e - 2\pi(e - 0) + 2\pi(e - 1) \quad \checkmark \\ &= \pi e - 2\pi e + 2\pi e - 2\pi \\ &= \pi e - 2\pi \quad \checkmark \end{aligned}$$

$$Q12)a) \int \frac{\sin^3 x}{\cos^2 x} dx$$

$$\text{let } u = \cos x \\ -du = \sin x dx$$

$$= - \int \frac{1-u^2}{u^2} du \quad \checkmark$$

$$= - \int u^{-2} - 1 du$$

$$= u^{-1} + u + C$$

$$= \sec x + \cos x + C \quad \checkmark$$

Alternatively

$$\int \frac{\sin^3 x}{\cos^2 x} dx = \int \sin x \tan^2 x dx$$

$$= \int \sin x (\sec^2 x - 1)$$

$$= \int \frac{\sin x}{\cos^2 x} - \sin x dx$$

$$= \int \sin x (\cos x)^{-2} - \sin x dx$$

$$= \sec x + \cos x + C$$

$$b) \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = -4x$$

$$\int \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = \int -4x dx$$

$$\frac{1}{2} v^2 = -2x^2 + C \quad \checkmark$$

When  $v = -2 \text{ m/s}$ ,  $x = 0$ .

$$\therefore C = 2$$

$$v^2 = -4x^2 + 4 \quad \checkmark$$

$$v = -\sqrt{4-4x^2} \quad \text{since } v = -2 \text{ when } x = 0.$$

$$= -2\sqrt{1-x^2} \quad \checkmark$$

c) Assume that  $\sqrt{11} - \sqrt{5} \geq \sqrt{2}$

$$(\sqrt{11} - \sqrt{5})^2 \geq 2$$

$$11 - 2\sqrt{55} + 5 \geq 2$$

$$2\sqrt{55} \leq 14$$

$$\sqrt{55} \leq 7$$

$$55 \leq 49 \quad (\text{which is incorrect}) \quad \checkmark$$

$\therefore$  By contradiction,  $\sqrt{11} - \sqrt{5} < \sqrt{2} \quad \checkmark$

d) Area of a quarter of a circle.

$$= \frac{1}{4} \times \pi \times 6^2 \quad \checkmark$$

$$= 9\pi \quad \checkmark$$

Alternatively

$$\text{let } x = 6 \sin \theta \\ dx = 6 \cos \theta d\theta$$

$$x = 0, \\ \theta = 0$$

$$x = 6, \\ \theta = \frac{\pi}{2}$$

$$= 6 \int_0^{\frac{\pi}{2}} \cos \theta \times 6 \cos \theta d\theta$$

$$= 36 \int_0^{\frac{\pi}{2}} \frac{\cos 2\theta + 1}{2} d\theta$$

$$= 18 \int_0^{\frac{\pi}{2}} \cos 2\theta + 1 d\theta$$

$$= 18 \left[ \frac{\sin 2\theta}{2} + \theta \right]_0^{\frac{\pi}{2}} \quad \checkmark$$

$$= 18 \left( 0 + \frac{\pi}{2} - 0 \right)$$

$$= 9\pi \quad \checkmark$$

e) let  $z = x + iy$

$$\operatorname{Re}\left(\frac{x + iy - ib}{x + iy - a}\right) = 0$$

$$\operatorname{Re}\left(\frac{x + i(y-b)}{(x-a) + iy} \times \frac{(x-a) - iy}{(x-a) - iy}\right) = 0$$

$$\operatorname{Re}\left(\frac{x(x-a) - ixy + i(y-b)(x-a) + y(y-b)}{(x-a)^2 + y^2}\right) = 0$$

$$\frac{x(x-a) + y(y-b)}{(x-a)^2 + y^2} = 0 \quad \checkmark$$

$$x(x-a) + y(y-b) = 0$$

$$x^2 - ax + y^2 - by = 0$$

$$\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{a^2}{4} + \frac{b^2}{4} \quad \checkmark$$

$$\text{centre } \left(\frac{a}{2}, \frac{b}{2}\right) \quad \text{radius } \sqrt{\frac{a^2 + b^2}{4}} \quad \checkmark$$

i)  $\vec{OP} = x\hat{i} + y\hat{j}$

$$\vec{OP} = (x-2a)\hat{i} + y\hat{j}$$

$$= x\hat{i} + \sqrt{a^2 - (x-a)^2}\hat{j} \quad \checkmark = (x-2a)\hat{i} + \sqrt{a^2 - (x-a)^2}\hat{j}$$

ii)  $\vec{OP} \cdot \vec{OP} = x(x-2a) + a^2 - (x-a)^2 \quad \checkmark$

$$= x^2 - 2ax + a^2 - x^2 + 2ax - a^2$$

$$= 0 \quad \checkmark$$

Q13) a) i)  $x = 2 + \sin 2t + \sqrt{3} \cos 2t$

$$\dot{x} = 2 \cos 2t - 2\sqrt{3} \sin 2t$$

$$\ddot{x} = -4 \sin 2t - 4\sqrt{3} \cos 2t \quad \checkmark$$

$$\ddot{x} = -4(\sin 2t + \sqrt{3} \cos 2t)$$

$$\ddot{x} = -4(x-2)$$

which is in the form of  $\ddot{x} = -n^2(x-a)$ . ✓

ii)  $a=2$  ✓

iii) Max speed occurs at the centre of motion.

$\therefore$  at  $x=2$

$$2 = 2 + \sin 2t + \sqrt{3} \cos 2t$$

$$\sin 2t + \sqrt{3} \cos 2t = 0$$

$$\tan \alpha = \sqrt{3} \quad r = \frac{\sqrt{1+3}}{2}$$

$$\alpha = \frac{\pi}{3}$$

$$\therefore \begin{cases} 2 \sin(2t + \frac{\pi}{3}) = 0 \\ \sin(2t + \frac{\pi}{3}) = 0 \end{cases} \quad \checkmark$$

$$2t + \frac{\pi}{3} = 0, \pi, \dots$$

$$t = \cancel{\frac{\pi}{6}}, \frac{\pi}{3}$$

$$\therefore t = \frac{\pi}{3} \quad \checkmark$$

b)  $(a^2-1) \geq 0$

$$a^4 - 2a^2 + 1 \geq 0$$

$$a^4 + a^2 + 1 \geq 3a^2$$

$$\therefore \frac{a^4 + a^2 + 1}{a^2} \geq 3 \quad \checkmark$$

$$\begin{aligned}
 \text{c) i) } \left( z - \frac{1}{z} \right)^5 &= z^5 - 5 \frac{z^4}{z} + 10 \frac{z^3}{z^2} - 10 \frac{z^2}{z^3} + 5 \frac{z}{z^4} - \frac{1}{z^5} \quad \checkmark \\
 &= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5} \\
 &= (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1}) \quad \checkmark
 \end{aligned}$$

$$\text{ii) let } z = \cos \theta + i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = \cos n\theta - i \sin n\theta$$

$$z^n - z^{-n} = 2i \sin n\theta$$

$$\text{From (i), } (z - z^{-1})^5 = (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$$

$$(2i \sin \theta)^5 = 2i \sin 5\theta - 5 \times 2i \sin 3\theta + 10 \times 2i \sin \theta \quad \checkmark$$

$$32i^5 \sin^5 \theta = i(2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta)$$

$$32i \sin^5 \theta = i(2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta)$$

$$32 \sin^5 \theta = 2 \sin 5\theta - 10 \sin 3\theta + 20 \sin \theta \quad \checkmark$$

$$\text{iii) } \int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta = \frac{1}{16} \int_0^{\frac{\pi}{2}} \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta \, d\theta$$

$$= \frac{1}{16} \left[ -\frac{1}{5} \cos 5\theta + \frac{5}{3} \cos 3\theta - 10 \cos \theta \right]_0^{\frac{\pi}{2}} \quad \checkmark$$

$$= \frac{1}{16} \left[ \left( -\frac{1}{5} \cos \frac{5\pi}{2} + \frac{5}{3} \cos \frac{3\pi}{2} - 10 \cos \frac{\pi}{2} \right) - \left( -\frac{1}{5} \times 1 + \frac{5}{3} \times 1 - 10 \times 1 \right) \right]$$

$$= \frac{1}{16} \left[ 0 - \left( -\frac{1}{5} + \frac{5}{3} - 10 \right) \right]$$

$$= \frac{8}{15} \quad \checkmark$$

d) let the consecutive numbers be  $k-1$ ,  $k$  and  $k+1$  for  $k \in \mathbb{Z}$

$$\begin{aligned}\therefore P &= (k-1) \times k \times (k+1) \\ &= k^3 - k\end{aligned}$$

Case 1:  $k$  is even

$$k = 2m, \quad m \in \mathbb{Z}$$

$$\begin{aligned}\therefore P &= (2m)^3 - 2m \\ &= 8m^3 - 2m \\ &= 2(4m^3 - m) \\ &= 2P, \quad P \in \mathbb{Z}\end{aligned}$$

$\therefore$  true if  $k$  is even

Case 2:  $k$  is odd

$$\text{let } k = 2m+1$$

$$\begin{aligned}\therefore P &= (2m+1)^3 - (2m+1) \\ &= 8m^3 + 12m^2 + 6m + 1 - 2m - 1 \\ &= 2(4m^3 + 6m^2 + 2m) \\ &= 2Q, \quad Q \in \mathbb{Z}\end{aligned}$$

$\therefore$  true if  $k$  is odd

$\therefore$  the product of three consecutive numbers is even.

$$Q14) a) i) \begin{matrix} P(-1, 2, 3) \\ Q(-2, 4, 2) \end{matrix}$$

$$\overrightarrow{PQ} = (-2 - (-1))\underline{i} + (4 - 2)\underline{j} + (2 - 3)\underline{k} \\ = -\underline{i} + 2\underline{j} - \underline{k}$$

$$r = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} \quad \checkmark$$

$$ii) \cos(\cos^{-1} \frac{1}{6}) = \frac{-1 \times -1 + 2 \times b + -1 \times 2}{\sqrt{b^2 + 5} \times \sqrt{6}}$$

$$\frac{2b - 1}{\sqrt{b^2 + 5} \times \sqrt{6}} = \frac{1}{6} \quad \checkmark$$

$$2b - 1 = \frac{1}{\sqrt{6}} \sqrt{b^2 + 5}$$

$$(2b - 1)^2 = \frac{1}{6} (b^2 + 5)$$

$$24b^2 - 24b + 6 = b^2 + 5$$

$$23b^2 - 24b + 1 = 0$$

$$(b - 1)(23b - 1) = 0$$

$b = 1$  , since  $b = \frac{1}{23}$  does not produce an acute angle when substituting  $\checkmark$

$$b) i) \text{ If } y > x, \text{ then } y^3 + yx^2 > x^3 + xy^2 \quad \checkmark$$

$$ii) \begin{matrix} y > x \\ y(x^2 + y^2) > x(x^2 + y^2) \\ y^3 + yx^2 > x^3 + xy^2 \end{matrix} \quad \checkmark$$

$\therefore$  the contrapositive is true.

$$\text{Hence, if } y \leq x, \text{ then } y^3 + yx^2 \leq x^3 + xy^2 \quad \checkmark$$

$$c) i) \quad \vec{OA} = \begin{pmatrix} -p \\ 5+4p \\ 4+3p \end{pmatrix} \quad \vec{OB} = \begin{pmatrix} -2-q \\ 4+2q \\ 1+2q \end{pmatrix}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= \begin{pmatrix} -2-q+p \\ -1+2q-4p \\ -3+2q-3p \end{pmatrix} \quad \checkmark$$

ii)  $|\vec{AB}|$  is a minimum when  $\vec{AB}$  is perpendicular to both lines.

$$\vec{AB} \cdot \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix} = 0$$

$$\begin{aligned} 2+q-p-4+8q-16p-9+6q-9p &= 0 \\ -11+15q-26p &= 0 \quad \textcircled{1} \end{aligned}$$

$$\vec{AB} \cdot \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} = 0$$

$$\begin{aligned} 2+q-p-2+4q-8p-6+4q-6p &= 0 \\ -6+9q-15p &= 0 \\ 3q &= 5p+2 \quad \textcircled{2} \end{aligned}$$

$$\begin{aligned} \text{sub } \textcircled{2} \text{ into } \textcircled{1}, \quad -11+5(5p+2)-26p &= 0 \\ -11+25p+10-26p &= 0 \\ p &= -1 \quad \text{sub into } \textcircled{2} \end{aligned}$$

$$\begin{aligned} 3q &= -5+2 \\ 3q &= -3 \\ q &= -1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \therefore \vec{AB} &= \begin{pmatrix} -2-(-1)-1 \\ -1+2(-1)-4(-1) \\ -3+2(-1)-3(-1) \end{pmatrix} \\ &= \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix} \quad \checkmark \end{aligned}$$

$$\begin{aligned} |\vec{AB}| &= \sqrt{(-2)^2 + 1^2 + (-2)^2} \\ &= 3 \text{ units} \quad \checkmark \end{aligned}$$

$$d) \text{ For } S_1, \quad x^2 - 2x + y^2 + 4y + z^2 - 4z = 0$$

$$x^2 - 2x + 1 + y^2 + 4y + 4 + z^2 - 4z + 4 = 9$$

$$(x-1)^2 + (y+2)^2 + (z-2)^2 = 9$$

$$\text{centre} = (-1, -2, -2) \quad \text{radius} = 3$$

$$\text{For } S_2, \quad x^2 + 10x + y^2 + z^2 + 2z = -10$$

$$x^2 + 10x + 100 + y^2 + z^2 + 2z + 1 = 16$$

$$(x+5)^2 + y^2 + (z+1)^2 = 16$$

$$\text{centre} = (-5, 0, -1) \quad \text{radius} = 4 \quad \checkmark$$

$$\text{The distance the centres} = \sqrt{(-5-1)^2 + (-2)^2 + (-1-2)^2}$$

$$= 7$$

The distance between the centres is equal to the sum of the radii. Therefore, the spheres intersect at a point.  $\checkmark$

$$e) \text{ let } u = e^x$$

$$du = e^x dx$$

$$= \int \frac{1}{u^2 + 3u + 2} du$$

$$= \int \frac{1}{(u+2)(u+1)} du \quad \checkmark \quad \frac{1}{(u+2)(u+1)} = \frac{a}{u+2} + \frac{b}{u+1}$$

$$1 = a(u+1) + b(u+2)$$

$$\text{let } u = -1, \quad u = -2,$$

$$b = 1 \quad a = -1$$

$$= \int \frac{-1}{u+2} + \frac{1}{u+1} du \quad \checkmark$$

$$= \ln|u+1| - \ln|u+2| + C$$

$$= \ln \left| \frac{e^x + 1}{e^x + 2} \right| + C \quad \checkmark$$

$$\begin{aligned}
 \text{Q15) a) } \int (1 + 2x^2)e^{x^2} dx &= \int e^{x^2} dx + \int 2x^2 e^{x^2} dx \\
 &= \int e^{x^2} dx + \int x \times 2x e^{x^2} dx \quad \checkmark \\
 &\quad \begin{array}{ll} u = x & v' = 2x e^{x^2} \\ u' = 1 & v = e^{x^2} \end{array} \\
 &= \int e^{x^2} dx + x e^{x^2} - \int e^{x^2} dx \\
 &= x e^{x^2} + c \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{b) i) } |\sin(\alpha + \beta)| &= |\sin \alpha \cos \beta + \cos \alpha \sin \beta| \\
 &\leq |\sin \alpha \cos \beta| + |\cos \alpha \sin \beta| \\
 &= |\sin \alpha| |\cos \beta| + |\cos \alpha| |\sin \beta| \\
 &\leq |\sin \alpha| + |\sin \beta| \quad \text{since } |\cos \theta| \leq 1 \quad \checkmark
 \end{aligned}$$

ii) Equality holds if one of  $\alpha$  or  $\beta$  is a multiple of  $\pi$ .  $\checkmark$

$$\text{c) i) } \ddot{x} = -k\dot{x}$$

$$\frac{d\dot{x}}{dt} = -k\dot{x}$$

$$\frac{dt}{d\dot{x}} = \frac{-1}{k\dot{x}}$$

$$t = \int \frac{-1}{k\dot{x}} dt$$

$$t = -\frac{1}{k} \ln(\dot{x}) + c$$

$$\dot{x} = u \cos \alpha \text{ when } t=0$$

$$\therefore c = \frac{1}{k} \ln(u \cos \alpha) \quad \checkmark$$

$$\therefore t = -\frac{1}{k} \ln(\dot{x}) + \frac{1}{k} \ln(u \cos \alpha)$$

$$kt = \ln\left(\frac{u \cos \alpha}{\dot{x}}\right)$$

$$\frac{u \cos \alpha}{\dot{x}} = e^{kt}$$

$$\therefore \dot{x} = u \cos \alpha e^{-kt} \quad \checkmark$$

$$\text{ii) } \dot{y} = \frac{1}{k} \left[ (k \sin \alpha + g) e^{-kt} - g \right]$$

$$\ddot{y} = -k \times \frac{1}{k} (k \sin \alpha + g) e^{-kt}$$

$$= -(k \sin \alpha + g) e^{-kt} \quad \checkmark$$

$$-k\dot{y} - g = -k \times \frac{1}{k} \left[ (k \sin \alpha + g) e^{-kt} - g \right] - g$$

$$= - \left[ (k \sin \alpha + g) e^{-kt} - g \right] - g$$

$$= -(k \sin \alpha + g) e^{-kt}$$

$$\therefore \ddot{y} = -k\dot{y} - g \quad \checkmark$$

Hence  $\dot{y} = \frac{1}{k} \left[ (k \sin \alpha + g) e^{-kt} - g \right]$  satisfies  $\ddot{y} = -k\dot{y} - g$

iii) At max height,  $\dot{y} = 0$

$$\therefore \frac{1}{k} \left[ (k \sin \alpha + g) e^{-kt} - g \right] = 0$$

$$(k \sin \alpha + g) e^{-kt} = g$$

$$e^{kt} = \frac{k \sin \alpha + g}{g}$$

$$\therefore t = \frac{1}{k} \ln \left( \frac{k \sin \alpha + g}{g} \right) \quad \checkmark$$

iv) From (i),  $\dot{x} = u \cos \alpha e^{-kt}$

$$x = \int u \cos \alpha e^{-kt} dt$$

$$= -\frac{u}{k} \cos \alpha e^{-kt} + C_2$$

$x = 0$  when  $t = 0$ ,

$$\therefore C_2 = \frac{u}{k} \cos \alpha$$

$$\therefore x = \frac{u \cos \alpha}{k} (1 - e^{-kt}) \quad \checkmark$$

As  $t \rightarrow \infty$ ,  $e^{-kt} \rightarrow 0$ .

The limiting value of the horizontal displacement is  $x = \frac{u \cos \alpha}{k} \quad \checkmark$

c)  $\vec{OP} = \vec{ON} + \vec{NP}$   
 $\vec{p} = \vec{ON} + \vec{MN}$   
 $= \vec{n} + \vec{n} - \vec{m}$   
 $= 2\vec{n} - \vec{m}$

$$\vec{OQ} = \vec{ON} + \vec{NQ}$$
  
 $\vec{q} = \vec{ON} + \lambda \vec{ON}$   
 $= (1 + \lambda) \vec{ON}$   
 $= (1 + \lambda) \vec{n} \quad \checkmark$

$$\vec{QP} = \vec{p} - \vec{q}$$
  
 $= 2\vec{n} - \vec{m} - (1 + \lambda) \vec{n}$   
 $= (1 - \lambda) \vec{n} - \vec{m} \quad \checkmark$

$$\vec{QP} \cdot \vec{MN} = 0$$
  
 $(\vec{m} - (1 - \lambda) \vec{n}) \cdot (\vec{n} - \vec{m}) = 0 \quad \checkmark$   
 $\vec{m} \cdot \vec{n} - \vec{m} \cdot \vec{m} - (1 - \lambda) \vec{n} \cdot \vec{n} + (1 - \lambda) \vec{n} \cdot \vec{m} = 0$   
 $(\lambda - 1) |\vec{n}|^2 + (2 - \lambda) \vec{m} \cdot \vec{n} - |\vec{m}|^2 = 0$   
 $\lambda (|\vec{n}|^2 - \vec{m} \cdot \vec{n}) + 2 \vec{m} \cdot \vec{n} - |\vec{m}|^2 - |\vec{n}|^2 = 0$   
 $\lambda (|\vec{n}|^2 - \vec{m} \cdot \vec{n}) = |\vec{m}|^2 + |\vec{n}|^2 - 2 \vec{m} \cdot \vec{n}$   
 $\lambda = \frac{|\vec{m}|^2 + |\vec{n}|^2 - 2 \vec{m} \cdot \vec{n}}{|\vec{n}|^2 - \vec{m} \cdot \vec{n}} \quad \checkmark$

$$\text{Q16) a) i) } \ddot{x} = g - kv \quad , \quad k > 0$$

At terminal velocity,  $\ddot{x} = 0$

$$\begin{aligned} \therefore g - kv &= 0 \\ g &= kv \\ v &= \frac{g}{k} \end{aligned}$$

$$\therefore \text{terminal velocity, } T = \frac{g}{k} \quad \checkmark$$

$$\text{ii) } \ddot{x} = g - kv$$

$$v \frac{dv}{dx} = g - kv$$

$$\frac{dv}{dx} = \frac{g - kv}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv}$$

$$= \frac{1}{k} \left( \frac{v}{\frac{g}{k} - v} \right)$$

$$= \frac{1}{k} \left( \frac{v}{T - v} \right) \quad \text{since } T = \frac{g}{k} \quad \checkmark$$

$$x = -\frac{1}{k} \int \frac{T - v}{T - v} - \frac{T}{T - v} dv$$

$$x = -\frac{1}{k} \int 1 - \frac{T}{T - v} dv$$

$$= -\frac{1}{k} \left[ v + T \ln(T - v) \right] + C$$

When  $x = 0$ ,  $v = 0$

$$\therefore C = \frac{T \ln T}{k}$$

$$\therefore x = -\frac{1}{k} \left[ v + T \ln(T-v) - T \ln T \right] \quad \checkmark$$

$$= -\frac{T}{k} \ln \left( \frac{T-v}{T} \right) - \frac{v}{k}$$

$$= \frac{T}{k} \ln \left( \frac{T}{T-v} \right) - \frac{v}{k}$$

$$\text{When } v = \frac{T}{2}, \quad x = \frac{T}{k} \ln 2 - \frac{T}{2k} \quad \checkmark$$

$$k = \frac{g}{T}$$

$$\therefore x = \frac{T^2}{g} \ln 2 - \frac{T^2}{2g} \quad \checkmark$$

Alternatively

$$v \frac{dv}{dx} = g - kv$$

$$\frac{dx}{dv} = \frac{v}{g - kv}$$

$$\int_0^x dx = \int_0^{\frac{T}{2}} \frac{v}{g - kv} dv$$

$$x = -\frac{1}{k} \int_0^{\frac{T}{2}} \frac{g - kv - g}{g - kv} dv \quad \checkmark$$

$$x = -\frac{1}{k} \int_0^{\frac{T}{2}} dv + \frac{1}{k} \int_0^{\frac{T}{2}} \frac{g}{g - kv} dv$$

$$x = -\frac{1}{k} \int_0^{\frac{T}{2}} dv - \frac{g}{k^2} \int_0^{\frac{T}{2}} \frac{-k}{g - kv} dv$$

$$x = -\frac{1}{k} \times \frac{T}{2} - \frac{g}{k^2} \left[ \ln |g - kv| \right]_0^{\frac{T}{2}} \quad \checkmark$$

$$x = -\frac{T}{2k} - \frac{g}{k^2} \ln \left| g - \frac{Tk}{2} \right| + \frac{g}{k^2} \ln g$$

$$x = \frac{-T}{2k} + \frac{g}{k^2} \ln \left| \frac{2g}{2g - Tk} \right|$$

$$\text{let } k = \frac{g}{T} \quad \checkmark$$

$$\therefore x = \frac{-T^2}{2g} + \frac{T^2}{g} \ln \left| \frac{2g}{2g - g} \right|$$

$$x = \frac{-T^2}{2g} + \frac{T^2}{g} \ln 2 \quad \checkmark$$

$$\begin{aligned} \text{b) } P(x) &= (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n) \\ \ln P(x) &= \ln [(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)] \quad \checkmark \\ \ln P(x) &= \ln(x - \alpha_1) + \ln(x - \alpha_2) + \ln(x - \alpha_3) + \dots + \ln(x - \alpha_n) \end{aligned}$$

differentiating both sides with respect to  $x$ ,

$$\frac{P'(x)}{P(x)} = \frac{1}{x - \alpha_1} + \frac{1}{x - \alpha_2} + \frac{1}{x - \alpha_3} + \dots + \frac{1}{x - \alpha_n}$$

$$P'(x) = P(x) \left[ \frac{1}{x - \alpha_1} + \frac{1}{x - \alpha_2} + \frac{1}{x - \alpha_3} + \dots + \frac{1}{x - \alpha_n} \right] \quad \checkmark$$

c) i) test  $n=1$

$$\begin{aligned} \text{LHS} &= \sum_{r=1}^1 \frac{1}{\sqrt{r}} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 2(\sqrt{1+1} - 1) \\ &= 2(\sqrt{2} - 1) \\ &\doteq 0.828 \end{aligned}$$

$$1 > 0.828$$

$\therefore$  true for  $n=1$ .

Assume true for  $n=k$ ,

$$\sum_{r=1}^k \frac{1}{\sqrt{r}} > 2(\sqrt{k+1} - 1) \quad \text{where } k \text{ is an int and } > 0 \quad \checkmark$$

Prove true for  $n=k+1$ ,

$$\text{i.e. prove } \sum_{r=1}^{k+1} \frac{1}{\sqrt{r}} > 2(\sqrt{k+2} - 1)$$

$$\text{LHS} = \sum_{r=1}^k \frac{1}{\sqrt{r}} + \frac{1}{\sqrt{k+1}}$$

$$> 2(\sqrt{k+1} - 1) + \frac{1}{\sqrt{k+1}} \quad (\text{from assumption}) \quad \checkmark$$

$$= 2\sqrt{k+1} - 2 + \frac{1}{\sqrt{k+1}}$$

$$= \frac{2\sqrt{k+1} \times \sqrt{k+1}}{\sqrt{k+1}} + \frac{1}{\sqrt{k+1}} - 2$$

$$= \frac{2(k+1) + 1}{\sqrt{k+1}} - 2$$

$$= \frac{2k+3}{\sqrt{k+1}} - 2 \quad \checkmark$$

$$> \frac{2\sqrt{(k+1)(k+2)}}{\sqrt{k+1}} - 2 \quad (\text{given}) \quad \checkmark$$

$$= \frac{2\sqrt{k+1} \times \sqrt{k+2}}{\sqrt{k+1}} - 2$$

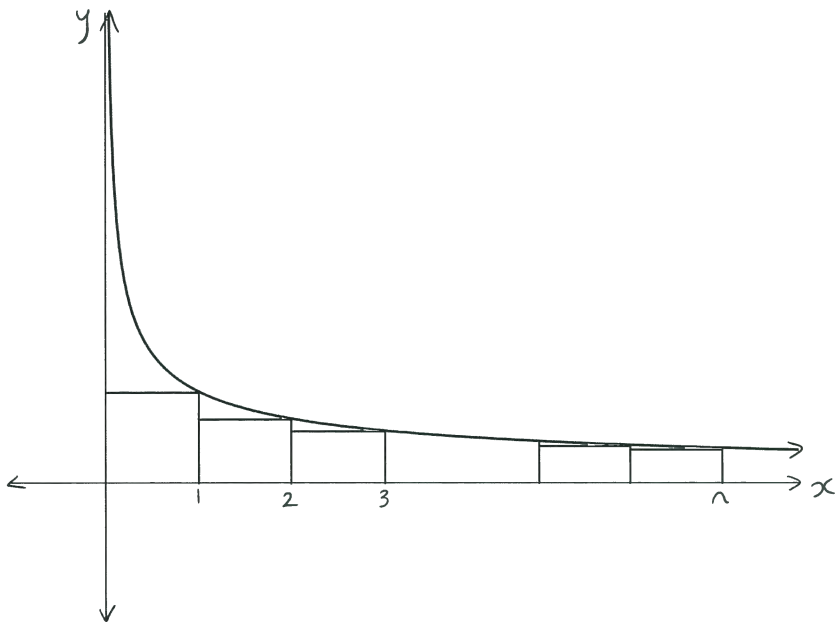
$$= 2\sqrt{k+2} - 2$$

$\therefore$  true for  $n=k+1$



$\therefore$  by induction, the result is true for all positive integers of  $n$ .

ii)



Exact area under the curve from 1 to  $n$  is  $\int_1^n \frac{1}{\sqrt{x}} dx$

Area of rectangles under the curve in the same interval =

$$1 \times \frac{1}{\sqrt{2}} + 1 \times \frac{1}{\sqrt{3}} + 1 \times \frac{1}{\sqrt{4}} + \dots + 1 \times \frac{1}{\sqrt{n}}$$

$$\therefore \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \dots + \frac{1}{\sqrt{n}} < \int_1^n \frac{1}{\sqrt{x}} dx \quad \checkmark$$

$$\therefore 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \dots + \frac{1}{\sqrt{n}} < \int_1^n \frac{1}{\sqrt{x}} dx + 1$$

$$\therefore \sum_{r=1}^n \frac{1}{\sqrt{r}} < \int_1^n \frac{1}{\sqrt{x}} dx + 1 \quad \checkmark$$

iii) From (i) and (ii),

$$2(\sqrt{n+1} - 1) < \sum_{r=1}^n \frac{1}{\sqrt{r}} < \int_1^n \frac{1}{\sqrt{x}} dx + 1 \quad \checkmark$$

let  $n = 10000$ ,

$$2(\sqrt{10001} - 1) < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < \left[ 2\sqrt{x} \right]_1^{10000} + 1$$

$$198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199 \quad \checkmark$$